Classical Mechanics

Lecture 7: Liouville's Theorem and Motion in a Magnetic Field

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Chapter 1

Liouville's Theorem and Motion in a Magnetic Field

Liouville's theorem is one of the central facts of Hamiltonian mechanics. It says that Hamiltonian evolution preserves volume in phase space. In quantum mechanics the corresponding idea is unitarity: distinguishable states do not quietly collapse into one another under the fundamental law of motion. In both theories the underlying issue is the conservation of information.

We shall first make this geometric statement precise and prove it directly from Hamilton's equations. We shall then use it to understand canonical area, chaotic stretching, coarse-graining, and entropy. In the second half we turn to a different problem, the motion of a charged particle in a magnetic field. That example introduces a velocity-dependent force while preserving the Lagrangian and Hamiltonian structures developed so far.

1.1 A Flow That Does Not Lose Information

A state of a classical system is a point in phase space. If we observe the system at equally spaced instants, the law of motion draws an arrow from each point to its successor. Determinism requires one outgoing arrow: the present state must determine what happens next. Reversibility also requires one incoming arrow: the present state must determine where it came from.

This already rules out two kinds of evolution. A trajectory cannot split into two futures, and two distinct trajectories cannot merge into a single state. Either event would destroy the one-to-one character of the microscopic law. For a finite set of states, a reversible update rule is therefore a permutation, decomposing the state space into disjoint cycles.

Hamiltonian mechanics makes a stronger statement. Fill a continuous phase space uniformly with imaginary dust. Each dust grain follows the local equations of motion. A small cloud of grains may stretch, bend, and shear, but its phase-space volume remains unchanged. The flow is incompressible.

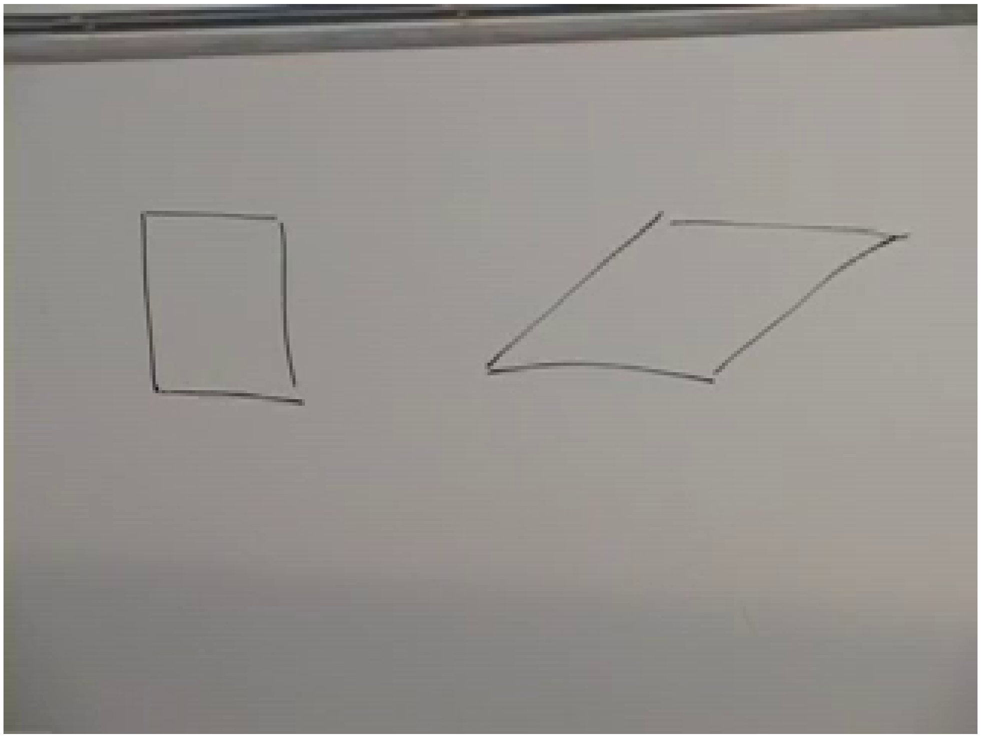

Figure 1.1: A square and its sheared image at 00:13:16. Hamiltonian flow need not preserve lengths or angles; the relevant invariant is phase-space area.

Question from the audience. Could a distribution of states converge or diverge even when each individual trajectory is deterministic?[a]

Response. For a general deterministic rule that possibility must be examined separately. Uniqueness of the future only forbids splitting. Liouville's theorem is stronger: for a Hamiltonian system, an entire region cannot be compressed into a smaller phase-space volume or expanded into a larger one.

[a]Lecture timestamp: 00:05:05–00:06:15.

There are several equivalent ways to picture the theorem. We may follow a small volume surrounding each point; we may follow an arbitrarily shaped region and compare its initial and final volumes; or we may begin with uniform dust and demand that its density remain uniform. These are local, regional, and density versions of the same incompressibility statement.

Volume preservation does not mean rigidity. A square can shear into a parallelogram, or a nearly round cloud can become a long, thin filament. The amount of phase space is fixed even though its shape and every ordinary distance inside it may change.

Question from the audience. If the initial region is connected, can the flow tear it apart or create a hole?[a]

Response. Not under a smooth Hamiltonian flow for which the solution exists uniquely. The time-evolution map and its inverse are continuous, so connectedness and topology are preserved. The region may become extremely distorted, but it is not cut and reattached.

[a]Lecture timestamp: 00:10:08–00:11:11.

Question from the audience. Can two points pass one another, and can the orientation of a little square change?[a]

Response. Their projections onto one coordinate axis may pass. What cannot happen is that two complete phase-space trajectories intersect at the same time and state. A little square may rotate or shear into a parallelogram; those changes are compatible with preserving its area.

[a]Lecture timestamp: 00:12:20–00:13:53.

Question from the audience. Does a chaotic double pendulum contradict this picture? Nearby initial conditions can separate rapidly.[a]

Response. No. Chaos concerns the growth of separations, not the growth of exact phase-space volume. Expansion along unstable directions is compensated by compression along other directions. A compact patch can become a complicated filament while retaining precisely the same volume.

[a]Lecture timestamp: 00:13:54–00:16:12.

For one canonical pair (q, p), volume means area. For N pairs it means the $2N$-dimensional volume element

$$d\Gamma = \prod_{i=1}^{N} dq_i \, dp_i. \tag{1.1}$$

Two points are close only when their complete states are close: their coordinates and their conjugate momenta must both be nearly equal.

Question from the audience. What exactly is the volume being preserved, and what does "nearby" mean?[a]

Response. For a single degree of freedom it is an area in the q-p plane; with more degrees of freedom it is the product of all the canonical coordinate and momentum intervals. Nearby states are close in this full space, not merely close in ordinary position.

[a]Lecture timestamp: 00:17:03–00:18:42.

Ordinary friction seems to violate the theorem. Many sliding states approach the same visible state of rest, and many throws of a die can settle on the same face. The apparent contraction arises because the description omits the table, the air, and the internal degrees of freedom. Mechanical energy becomes heat, sound, and microscopic motion. Information is transferred into those unobserved variables. A reduced phase space may contract even though the larger closed system remains Hamiltonian.

Question from the audience. When a die or a sliding block comes to rest, have many states not genuinely merged?[a]

Response. They have merged only in a coarse description that records the visible object and ignores its environment. Include the molecular state of the table, the air, and the emitted sound, and the final microscopic states remain distinct. The situation is analogous to energy: the block loses kinetic energy while the table warms by a correspondingly tiny amount.

[a]Lecture timestamp: 00:18:52–00:22:18.

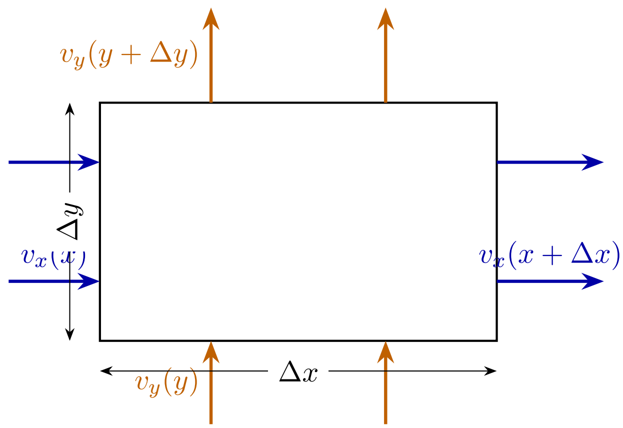

Figure 1.2: Instantaneous flux through a small rectangle. Incompressibility requires the sum of the two first-order outward-flux differences to vanish.

1.2 Incompressible Flow and the Divergence

Before proving the theorem in phase space, consider an ordinary fluid of points. In one dimension, let $v_x(x)$ be the instantaneous velocity. A small interval keeps a uniform density only if the flux entering one end equals the flux leaving the other. Thus

$$\frac{\partial v_x}{\partial x} = 0. \tag{1.2}$$

In one dimension an incompressible instantaneous flow is locally a rigid translation.

In two dimensions take a small rectangle with sides Δx and Δy. The difference between outward fluxes through the right and left faces is, to first order,

$$\frac{\partial v_x}{\partial x}\,\Delta x\,\Delta y. \tag{1.3}$$

The corresponding difference through the upper and lower faces is

$$\frac{\partial v_y}{\partial y}\,\Delta x\,\Delta y. \tag{1.4}$$

After dividing by the area, incompressibility becomes

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0, \qquad \text{or} \qquad \boldsymbol{\nabla}\cdot\boldsymbol{v} = 0. \tag{1.5}$$

This does not require the velocity to be the same everywhere. An increase of the x-velocity along x can be balanced by a decrease of the y-velocity along y. The rectangle may stretch in one direction and contract in the other.

Question from the audience. A point entering through one side may turn and later leave through another. Does the flux argument overlook that motion? [a]

Response. The argument is instantaneous. Take a snapshot and ask for the present rate of change of the number of points inside the rectangle. A later turn belongs to a later snapshot. Summing the flux through all faces at the same instant gives the local density change.

[a] Lecture timestamp: 00:37:09–00:37:49.

The same reasoning works in any number of dimensions:

$$\sum_{a=1}^{D} \frac{\partial v_a}{\partial x_a} = 0. \tag{1.6}$$

For a nonuniform density ρ, the general conservation law is the continuity equation

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot (\rho \boldsymbol{v}) = 0. \tag{1.7}$$

Uniform dust reduces this to zero divergence.

Question from the audience. Does the argument depend on two dimensions or on a uniform density?[a]

Response. The divergence formula extends to any number of dimensions. Uniform density is the convenient case used here. For a varying density one conserves the flux $\rho \boldsymbol{v}$, as in Eq. (1.7).

[a]Lecture timestamp: 00:37:50–00:40:07.

1.3 Proof of Liouville's Theorem

Return now to a phase space with canonical coordinates $(q_1, \ldots, q_N, p_1, \ldots, p_N)$. Hamilton's equations are

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}. \tag{1.8}$$

They define a phase-space velocity field

$$\boldsymbol{u}_H = (\dot{q}_1, \ldots, \dot{q}_N, \dot{p}_1, \ldots, \dot{p}_N). \tag{1.9}$$

Its divergence is

$$\begin{aligned} \boldsymbol{\nabla}_{q,p} \cdot \boldsymbol{u}_H &= \sum_{i=1}^{N} \left(\frac{\partial \dot{q}_i}{\partial q_i} + \frac{\partial \dot{p}_i}{\partial p_i} \right) \\ &= \sum_{i=1}^{N} \left(\frac{\partial^2 H}{\partial q_i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q_i} \right) = 0. \end{aligned} \tag{1.10}$$

Each pair of mixed derivatives cancels. This local identity is the whole proof.

Theorem 1.1 (Liouville's theorem)**.** *For a smooth Hamiltonian system, the time evolution preserves the canonical phase-space volume element* $d\Gamma = \prod_i dq_i \, dp_i$.

Proof. Equation (1.10) says that the Hamiltonian velocity field has zero divergence. The continuity equation then transports an initially uniform density without changing it. Equivalently, the Jacobian of the time-evolution map has unit determinant, so every transported region keeps its phase-space volume. □

This is the promised strengthening of reversibility. A one-to-one flow need not preserve volume in an arbitrary set of coordinates, but a Hamiltonian flow in canonical coordinates does so because of the crossed derivatives and the minus sign in Hamilton's equations.

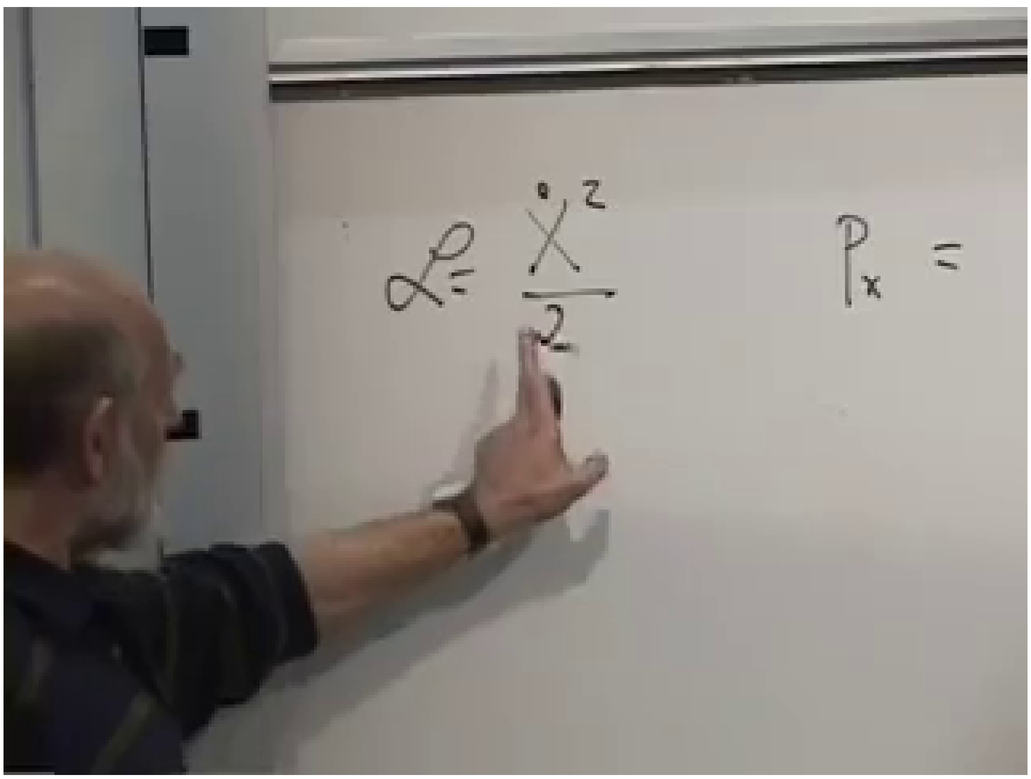

Figure 1.3: The mixed-derivative cancellation at 00:43:55. The q_i and p_i contributions enter with opposite signs.

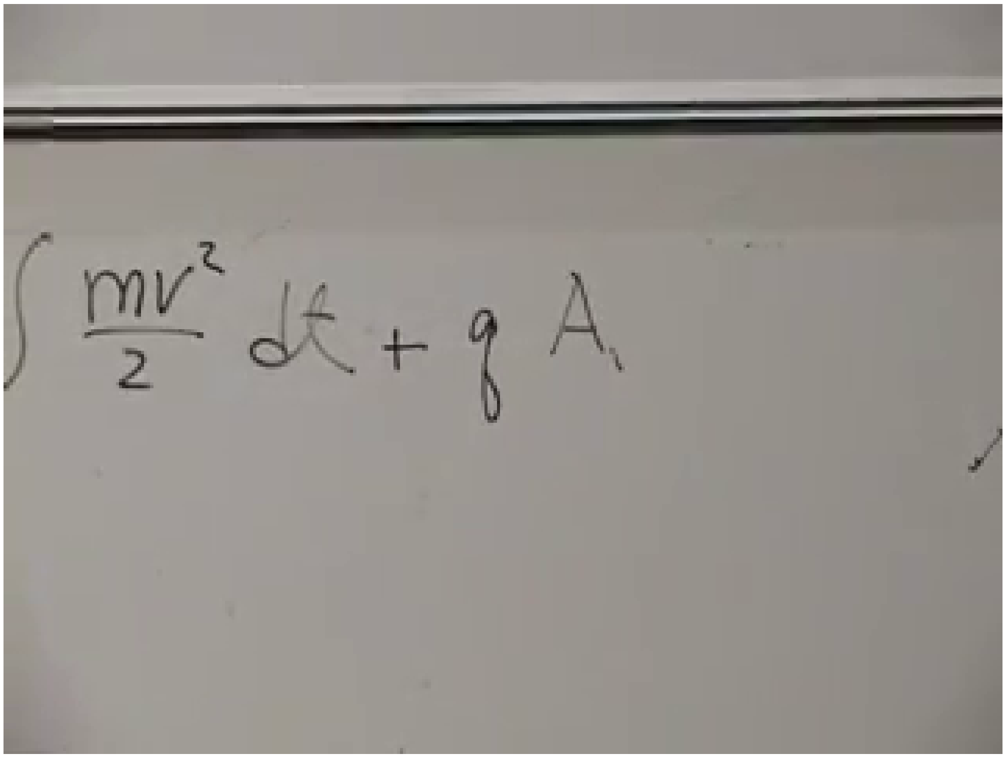

Figure 1.4: The reciprocal scaling of coordinate and canonical momentum at 00:49:45: $y = \alpha x$ while $p_y = p_x/\alpha$.

1.4 Canonical Area, Not Distance

For one degree of freedom, Liouville volume is area in the q-p plane. A simple change of coordinate shows why neither factor is separately invariant. Take a free particle of unit mass,

$$L = \frac{\dot{x}^2}{2}, \qquad p_x = \frac{\partial L}{\partial \dot{x}} = \dot{x}. \tag{1.11}$$

Introduce a rescaled coordinate

$$y = \alpha x, \qquad x = \frac{y}{\alpha}, \qquad \dot{x} = \frac{\dot{y}}{\alpha}. \tag{1.12}$$

Then

$$L = \frac{\dot{y}^2}{2\alpha^2}, \qquad p_y = \frac{\partial L}{\partial \dot{y}} = \frac{\dot{y}}{\alpha^2} = \frac{p_x}{\alpha}. \tag{1.13}$$

The coordinate interval stretches while the conjugate momentum interval shrinks:

$$\Delta y \, \Delta p_y = (\alpha \Delta x) \left(\frac{\Delta p_x}{\alpha} \right) = \Delta x \, \Delta p_x. \tag{1.14}$$

This transformation is canonical. It changes the appearance of a cell but not its symplectic area.

Question from the audience. Is α a time-evolution parameter, and does every coordinate transformation preserve area?[a]

Response. Here α is only a fixed change of scale, such as a change of units. The momentum must be transformed as the variable canonically conjugate to the new coordinate. An arbitrary independent reshuffling of q and p need not preserve area; a canonical transformation does.

[a]Lecture timestamp: 00:54:24–00:55:14.

Hamiltonian time evolution is itself a canonical transformation. It can move and deform a phase-space cell while preserving the canonical area element. That statement is entirely compatible with energy conversion. A falling ball trades potential energy for kinetic energy, so its position and momentum both change while its total energy and the area of an ensemble are preserved.

Question from the audience. Does a falling ball violate conservation because its momentum grows?[a]

Response. No. Potential energy decreases while kinetic energy increases. Liouville's theorem concerns the area occupied by a collection of nearby states, not the constancy of either coordinate or momentum for one state.

[a]Lecture timestamp: 00:53:27–00:54:23.

The same area already foreshadows quantum mechanics. Planck's constant has the dimensions of position times momentum, and the uncertainty principle sets a minimum scale for a quantum phase-space cell. The classical theorem does not impose that minimum; it supplies the canonical geometry in which the quantum statement will live.

Question from the audience. Why is Planck's constant associated with phase-space area rather than with one axis?[a]

Response. Because its dimensions are action, qp, and the uncertainty relation constrains the product of conjugate uncertainties. A rescaling can enlarge q while shrinking p, but their canonical product is the invariant quantity.

[a]Lecture timestamp: 00:52:26–00:52:55.

1.5 Chaos, Probability, and Coarse-Graining

Liouville's theorem does not say that a phase-space patch remains simple. In a chaotic system the patch may stretch exponentially, fold, and return across the same macroscopic region in thinner and thinner filaments. Exact volume is preserved throughout this process.

Phase-space area also supplies a natural invariant measure. Suppose a chaotic trajectory explores an accessible component of an energy surface and define probability by the fraction of a long observation time spent in a region. When the motion is ergodic with respect to the Liouville measure, relative occupation probabilities are proportional to invariant phase-space volume:

$$\frac{P(\Omega_1)}{P(\Omega_2)} = \frac{\mu_L(\Omega_1)}{\mu_L(\Omega_2)}. \tag{1.15}$$

Chaos by itself does not prove ergodicity, but it explains why this statistical description can become useful even when the microscopic equations are fully deterministic.

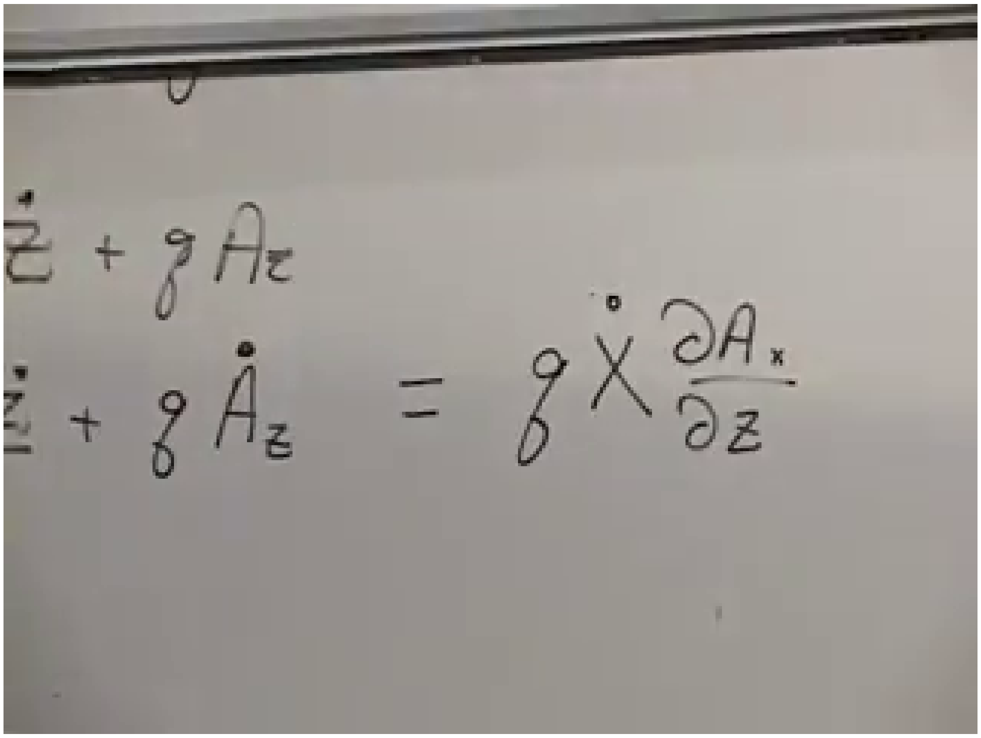

Figure 1.5: A compact patch and its filamented image at 01:03:40. The drawing illustrates deformation under chaotic evolution, not an increase of exact phase-space volume.

Question from the audience. Before quantum mechanics, what physical meaning does the phase-space area have?[a]

Response. It supplies the invariant measure used in statistical mechanics. Under the appropriate ergodic assumption, equal Liouville volumes carry equal long-time statistical weight, so relative areas become relative occupation probabilities.

[a]Lecture timestamp: 00:57:50–00:59:43.

Question from the audience. Is this probability stochastic, or is chaotic mechanics still deterministic?[a]

Response. The microscopic equations remain deterministic. The statistical description enters because finite precision cannot track the ever-finer distinctions among nearby initial conditions. Probability records long-time occupation or incomplete knowledge, not an indeterministic force inserted into the equations.

[a]Lecture timestamp: 00:59:44–01:02:14.

To see the role of finite precision, partition phase space into cells of the smallest size an apparatus can distinguish. These cells are fixed by the measurement convention; they are not carried along and distorted with the Hamiltonian flow. An initially compact patch may fit into a small set of cells. After it has become finely filamented, it intersects many more fixed cells.

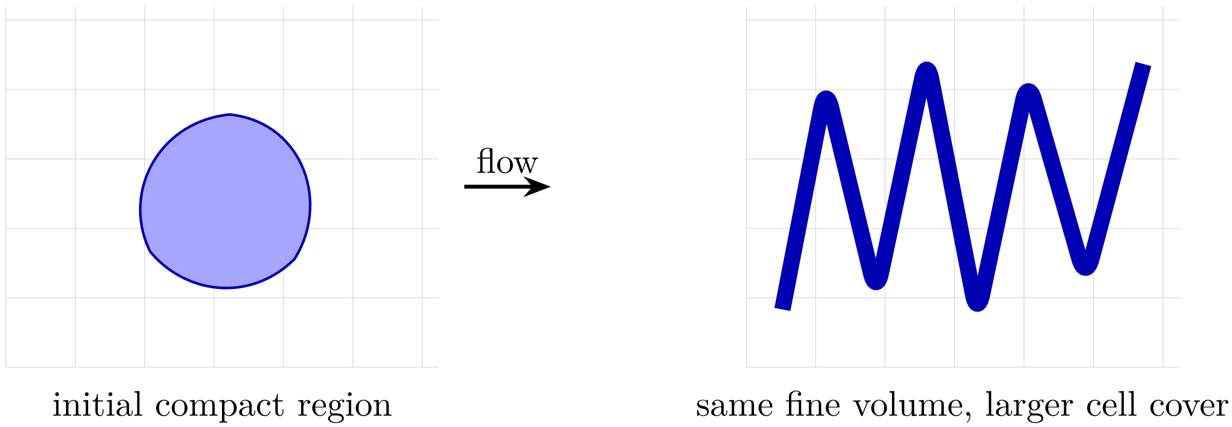

Figure 1.6: Fine-grained volume and coarse-grained coverage. Stretching and folding preserve exact volume, while finite-resolution cells can report a larger occupied region.

Question from the audience. Are the later coarse-graining cells the images of the initial cells under time evolution?[a]

Response. No. The coarse cells represent what the apparatus can distinguish at each time. If they were transported with the exact flow, they would stretch into the same fine filaments and no coarse growth would appear. The physically relevant comparison uses the same finite resolution before and after evolution.

[a]Lecture timestamp: 01:06:18–01:07:24.

The exact set still has its original volume. With infinitely precise calculation and measurement, no expansion would be found. With fixed finite resolution, however, the union of all cells intersecting the filaments can be much larger. Looking with "fuzzy eyes" fills in separations too fine to resolve.

Question from the audience. In what sense, then, can the phase-space area increase? [a]

Response. Only the coarse-grained area increases. The fine-grained set generated by Hamilton's equations retains exactly its original volume. The larger value counts every resolution cell touched by the thin filaments, including portions of those cells that the exact set does not occupy.

[a]Lecture timestamp: 01:07:39–01:11:26.

This distinction points toward the second law of thermodynamics. Entropy measures the logarithm of the detectable phase-space volume compatible with our macroscopic knowledge. With a reference volume Ω_0, one may write

$$S = k_B \log\left(\frac{\Omega_{\mathrm{cg}}}{\Omega_0}\right). \tag{1.16}$$

Microscopic information has not been destroyed. It has been hidden in correlations and filaments finer than the available resolution.

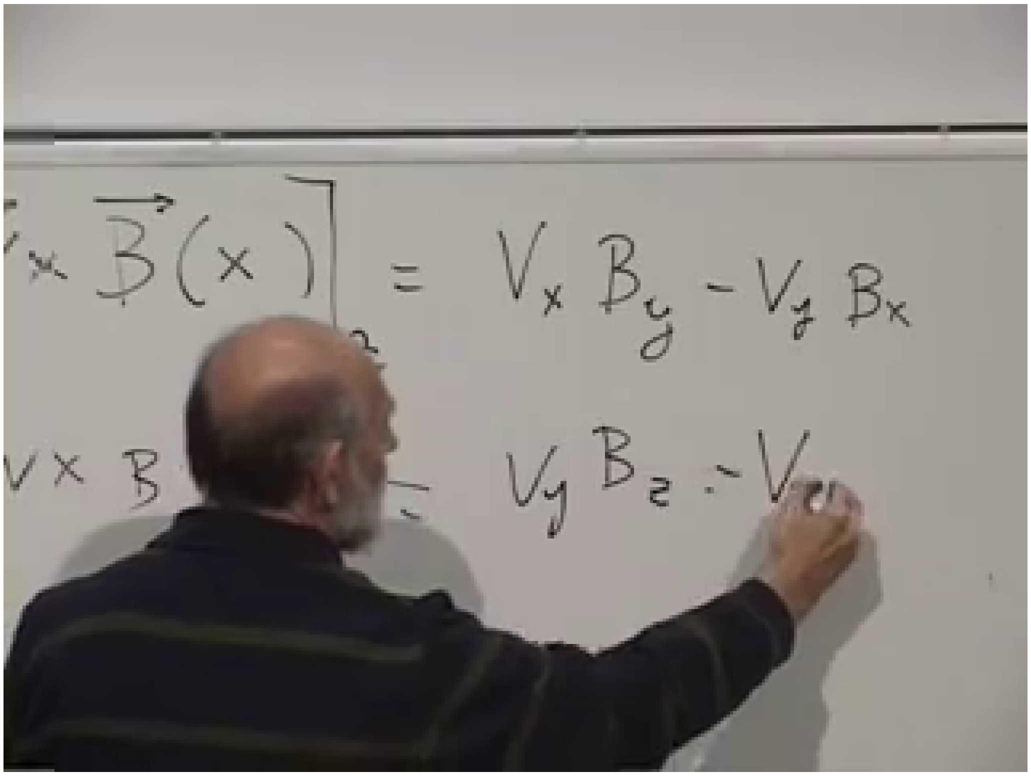

Figure 1.7: Components of $\boldsymbol{v} \times \boldsymbol{B}$ at 01:18:40. The upper line isolates the z-component used in the derivation below.

1.6 A Velocity-Dependent Fundamental Force

We now turn to the motion of a particle of mass m and charge q in a prescribed magnetic field. Its force is

$$\boxed{\boldsymbol{F} = q\,\boldsymbol{v} \times \boldsymbol{B}.} \tag{1.17}$$

Unlike the forces considered earlier, this one depends on velocity. Friction also depends on velocity, but friction is an effective dissipative force in a reduced description. The magnetic force is fundamental within the present approximation and follows from an action principle.

The cross product has components

$$\begin{aligned}
(\boldsymbol{v} \times \boldsymbol{B})_x &= v_y B_z - v_z B_y, \\
(\boldsymbol{v} \times \boldsymbol{B})_y &= v_z B_x - v_x B_z, \\
(\boldsymbol{v} \times \boldsymbol{B})_z &= v_x B_y - v_y B_x.
\end{aligned} \tag{1.18}$$

To construct a Lagrangian we introduce the vector potential $\boldsymbol{A}(\boldsymbol{x})$, defined by

$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}. \tag{1.19}$$

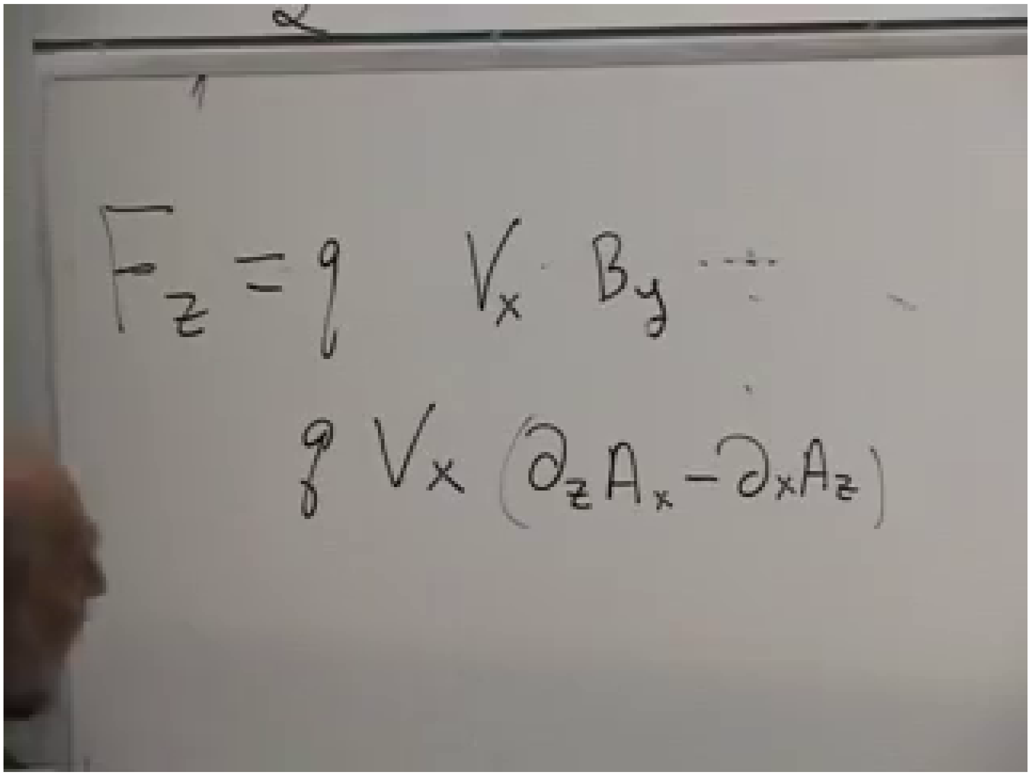

Figure 1.8: The z-component of the magnetic force rewritten with derivatives of the vector potential at 01:24:40.

In components,

$$
\begin{aligned}
B_x &= \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}, \\
B_y &= \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}, \\
B_z &= \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}.
\end{aligned}
\tag{1.20}
$$

Writing a field as a curl automatically gives $\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$, the absence of magnetic monopole sources in the classical theory being used here.

The z-component of the magnetic force can now be expressed directly in terms of $\boldsymbol{A}$:

$$
\begin{aligned}
F_z &= q(v_x B_y - v_y B_x) \\
&= q v_x \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + q v_y \left(\frac{\partial A_y}{\partial z} - \frac{\partial A_z}{\partial y} \right).
\end{aligned}
\tag{1.21}
$$

Question from the audience. Why introduce $\boldsymbol{A}$ when Newton's equation can be written directly with $\boldsymbol{B}$?[a]

Response. The magnetic field is sufficient for the force law, but we want an action, canonical momenta, and a Hamiltonian. The vector potential is the quantity that couples locally to the velocity in the Lagrangian. Its curl then makes the equations of motion depend on $\boldsymbol{B}$.

[a] Lecture timestamp: 01:19:06–01:21:15.

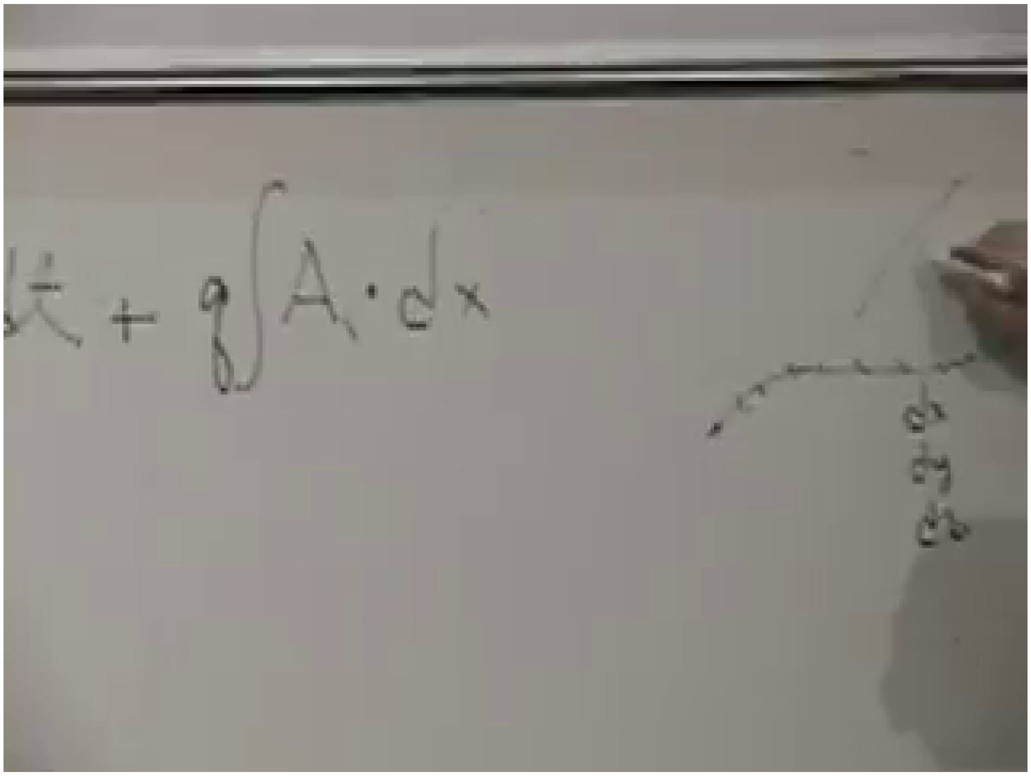

Figure 1.9: The action as a kinetic term plus a line integral of the vector potential at 01:28:50. The small path sketch emphasizes that the second term depends on the route through space.

1.7 The Magnetic Action and Canonical Momentum

The free-particle action is supplemented by a line integral along the particle's path:

$$S = \int \frac{1}{2} m v^2 \, dt + q \int_{\gamma} \boldsymbol{A} \cdot d\boldsymbol{x}. \tag{1.22}$$

Since $d\boldsymbol{x} = \dot{\boldsymbol{x}}\, dt$, the action takes the ordinary time-integral form

$$S = \int L \, dt, \qquad L = \frac{1}{2} m \dot{\boldsymbol{x}}^2 + q \, \boldsymbol{A}(\boldsymbol{x}) \cdot \dot{\boldsymbol{x}}. \tag{1.23}$$

We use units in which the coupling is written $q\boldsymbol{A} \cdot \boldsymbol{v}$. Other electromagnetic unit conventions may place a factor of $1/c$ in this term.

Question from the audience. How can the two terms in the action have the same units? [a]
Response. The action has units of energy times time. In the chosen convention $q\boldsymbol{A}$ has units of momentum, so $q\boldsymbol{A} \cdot d\boldsymbol{x}$ has units of action. A different convention may move a factor of c into the definition of the coupling, without changing the mechanics.

[a] Lecture timestamp: 01:29:51–01:31:17.

The canonical momentum is no longer simply the mechanical momentum:

$$p_i = \frac{\partial L}{\partial \dot{x}_i} = m\dot{x}_i + qA_i. \tag{1.24}$$

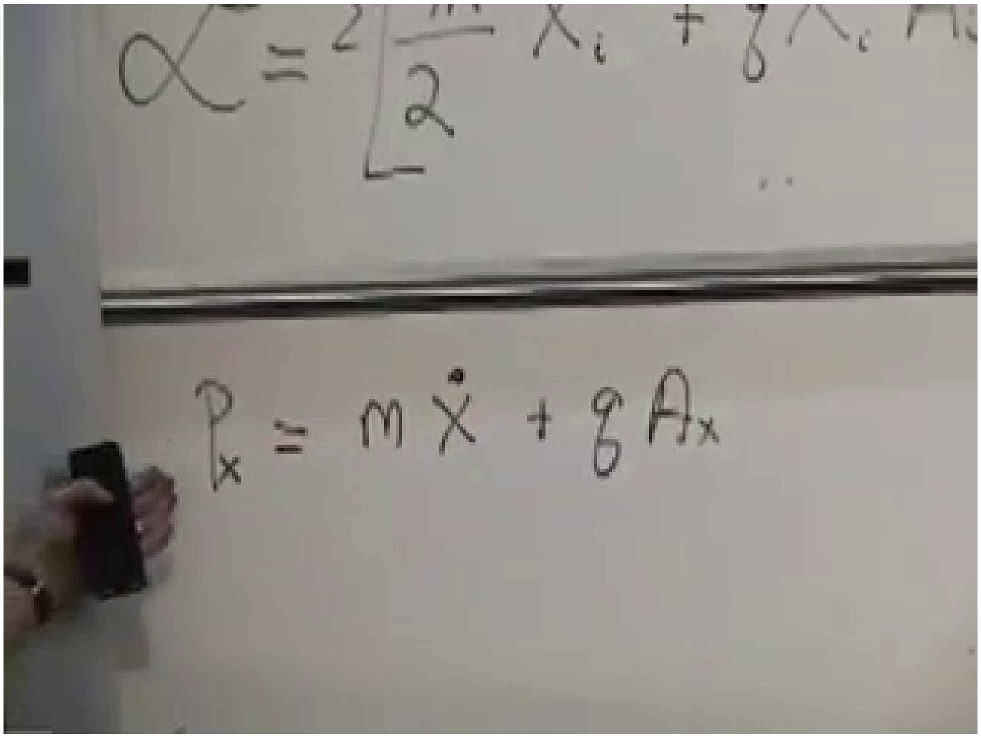

Figure 1.10: Canonical momentum in the x-direction at 01:35:20: $p_x = m\dot{x} + qA_x$.

Thus

$$\boldsymbol{p}_{\text{mech}} = m\boldsymbol{v} = \boldsymbol{p} - q\boldsymbol{A}. \tag{1.25}$$

This distinction is essential. Hamilton's equations use the canonical momentum $\boldsymbol{p}$; kinetic energy and the Lorentz force are most naturally expressed through the mechanical combination $\boldsymbol{p} - q\boldsymbol{A}$.

1.8 Recovering the Lorentz Force

We must verify that the proposed Lagrangian gives the correct equation of motion. Work out the z-component explicitly. The Euler–Lagrange equation is

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{z}} = \frac{\partial L}{\partial z}. \tag{1.26}$$

The left-hand side is

$$\frac{d}{dt}(m\dot{z} + qA_z) = m\ddot{z} + q\frac{dA_z}{dt}. \tag{1.27}$$

The right-hand side receives contributions from all three components of the vector potential:

$$\frac{\partial L}{\partial z} = q\dot{x}\frac{\partial A_x}{\partial z} + q\dot{y}\frac{\partial A_y}{\partial z} + q\dot{z}\frac{\partial A_z}{\partial z}. \tag{1.28}$$

The field is assumed static, meaning it has no explicit time dependence. It may still vary from point to point. Because the particle moves through that spatially varying field,

$$\frac{dA_z}{dt} = \dot{x}\frac{\partial A_z}{\partial x} + \dot{y}\frac{\partial A_z}{\partial y} + \dot{z}\frac{\partial A_z}{\partial z}. \tag{1.29}$$

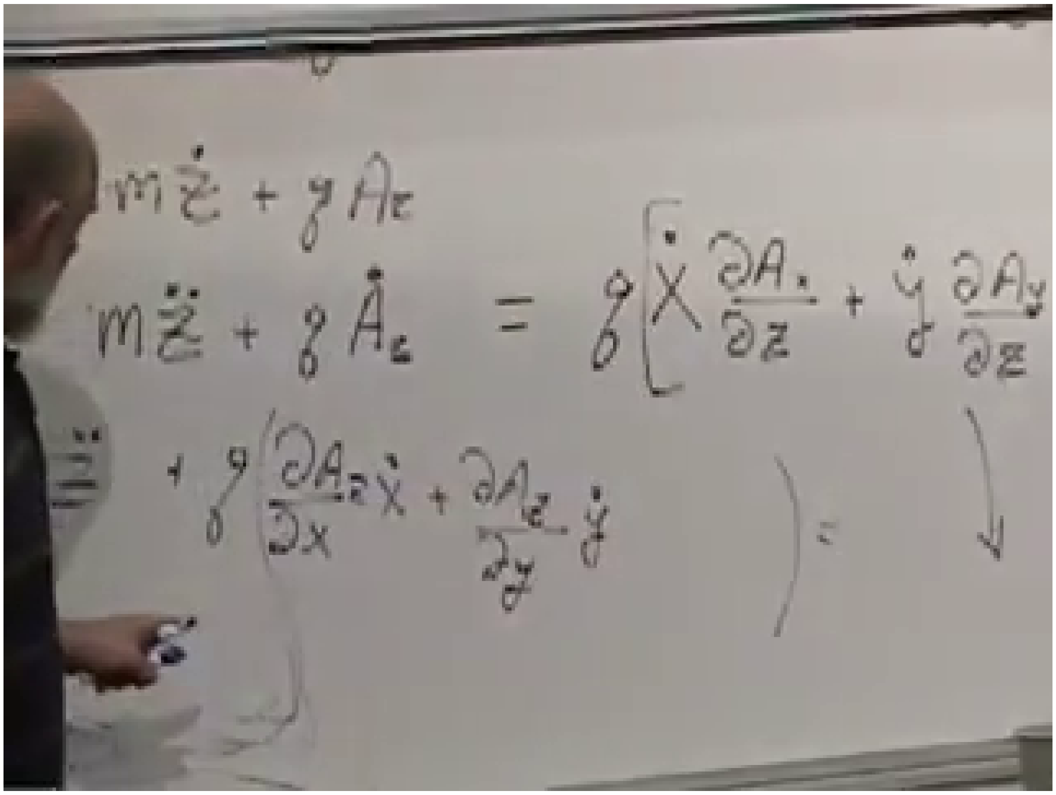

Figure 1.11: The two sides of the z-component Euler–Lagrange equation at 01:41:50. Terms proportional to $\dot z\,\partial A_z/\partial z$ occur on both sides and cancel.

Question from the audience. If the magnetic field is static, why is dA_z/dt not zero?[a]

Response. Static means no explicit dependence on time, not uniform in space. The particle samples different values of $\boldsymbol{A}(\boldsymbol{x})$ as it moves, so the value of A_z along the trajectory changes by the spatial chain rule.

[a] Lecture timestamp: 01:39:13–01:39:53.

Substitute Eq. (1.29) into the left-hand side of Eq. (1.26). The terms $q\dot z\,\partial A_z/\partial z$ cancel, leaving

$$\begin{aligned} m\ddot z &= q\dot x\left(\frac{\partial A_x}{\partial z}-\frac{\partial A_z}{\partial x}\right)+q\dot y\left(\frac{\partial A_y}{\partial z}-\frac{\partial A_z}{\partial y}\right)\\ &= q(v_xB_y-v_yB_x)=q(\boldsymbol{v}\times\boldsymbol{B})_z. \end{aligned} \tag{1.30}$$

The other components follow by cyclic permutation, and therefore

$$m\ddot{\boldsymbol{x}} = q\,\dot{\boldsymbol{x}}\times\boldsymbol{B}. \tag{1.31}$$

Question from the audience. Is the vector potential uniquely determined by the magnetic field?[a]

Response. No. For a static gauge transformation $\boldsymbol{A}\mapsto\boldsymbol{A}+\boldsymbol{\nabla}\chi$, the curl is unchanged. The Lagrangian changes by the total derivative $q\,d\chi/dt$, which changes the action only by endpoint terms and leaves the equations of motion unchanged.

[a] Lecture timestamp: 01:44:34–01:45:10.

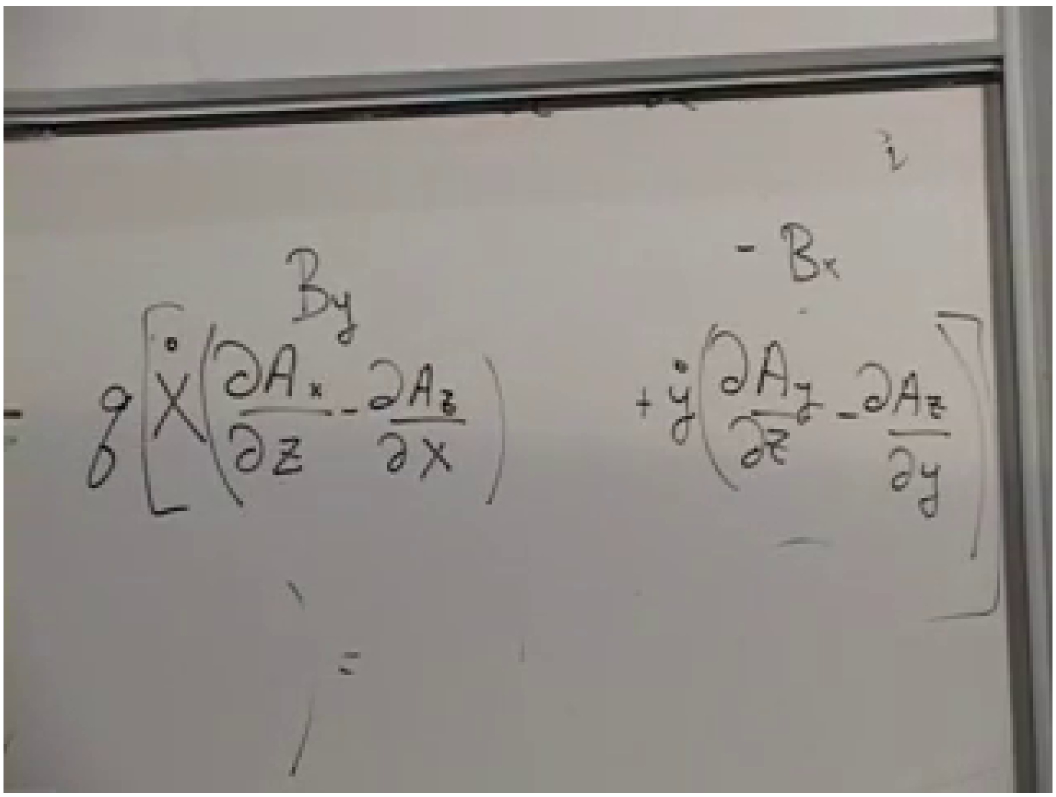

Figure 1.12: The surviving derivative combinations at 01:44:20, identified as B_y and $-B_x$, complete the z-component of the Lorentz force.

The contrast with friction is now sharp. A frictional force has a component opposite the velocity and changes the speed. A magnetic force is perpendicular to both $\boldsymbol{v}$ and $\boldsymbol{B}$:

$$\boldsymbol{F} \cdot \boldsymbol{v} = q(\boldsymbol{v} \times \boldsymbol{B}) \cdot \boldsymbol{v} = 0. \tag{1.32}$$

It changes the direction of motion without changing the speed.

Question from the audience. Why is a velocity-dependent magnetic force so different from friction?[a]

Response. The decisive issue is direction. Friction acts along the velocity axis and therefore changes kinetic energy. The magnetic force is perpendicular to the velocity, so its instantaneous power vanishes. It bends the trajectory but does not speed the particle up or slow it down.

[a]Lecture timestamp: 01:45:15–01:46:28.

1.9 Energy and the Hamiltonian

Assume that $\boldsymbol{A}$ has no explicit time dependence. The Hamiltonian is

$$H = \sum_i p_i \dot{x}_i - L. \tag{1.33}$$

First express it in terms of velocities:

$$\begin{aligned} H &= \sum_i (m\dot{x}_i + qA_i)\dot{x}_i - \left(\frac{1}{2}m\sum_i \dot{x}_i^2 + q\sum_i A_i\dot{x}_i\right) \\ &= \frac{1}{2}m\sum_i \dot{x}_i^2 = \frac{1}{2}mv^2. \end{aligned} \tag{1.34}$$

The two copies of the magnetic coupling cancel.

This agrees with the force argument: a static magnetic field does no work. In a uniform field perpendicular to the velocity, the particle moves in a circle at constant speed. Its direction continually changes, so it has an acceleration, but its kinetic energy remains constant.

To use Hamilton's equations we must eliminate the velocities. From Eq. (1.24),

$$\dot{x}_i = \frac{p_i - qA_i}{m}. \tag{1.35}$$

The Hamiltonian in canonical variables is therefore

$$\boxed{H(\boldsymbol{x}, \boldsymbol{p}) = \frac{1}{2m}\big(\boldsymbol{p} - q\boldsymbol{A}(\boldsymbol{x})\big)^2.} \tag{1.36}$$

It looks more complicated than the kinetic energy because the canonical momentum contains the vector potential. The combination $\boldsymbol{p} - q\boldsymbol{A} = m\boldsymbol{v}$ is the mechanical momentum.

Question from the audience. An accelerated charge radiates. How can its kinetic energy remain constant?[a]

Response. Radiation is being neglected in this particle-in-an-external-field model. A complete treatment must include the electromagnetic field as a dynamical Hamiltonian system carrying its own energy. Radiation reaction is a higher-order correction to the approximation used here.

[a]Lecture timestamp: 01:57:29–01:58:29.

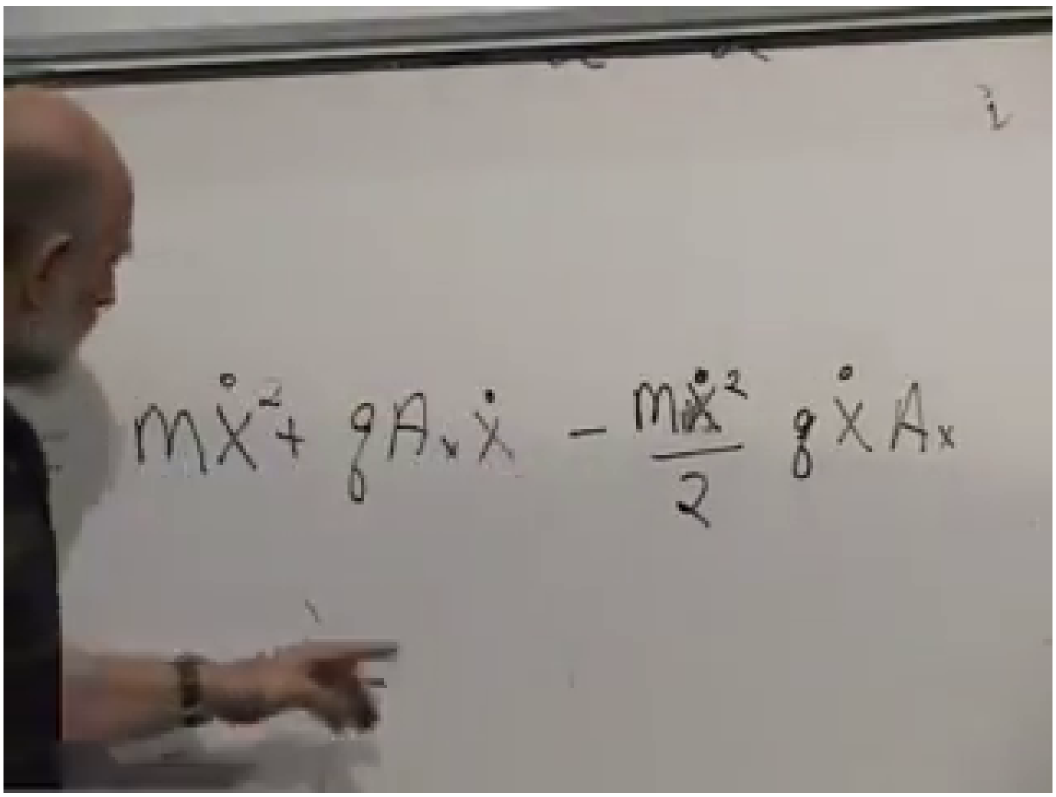

Figure 1.13: The x-component of the Legendre transform at 01:54:20. The $qA_x\dot{x}$ terms cancel, leaving $m\dot{x}^2/2$.

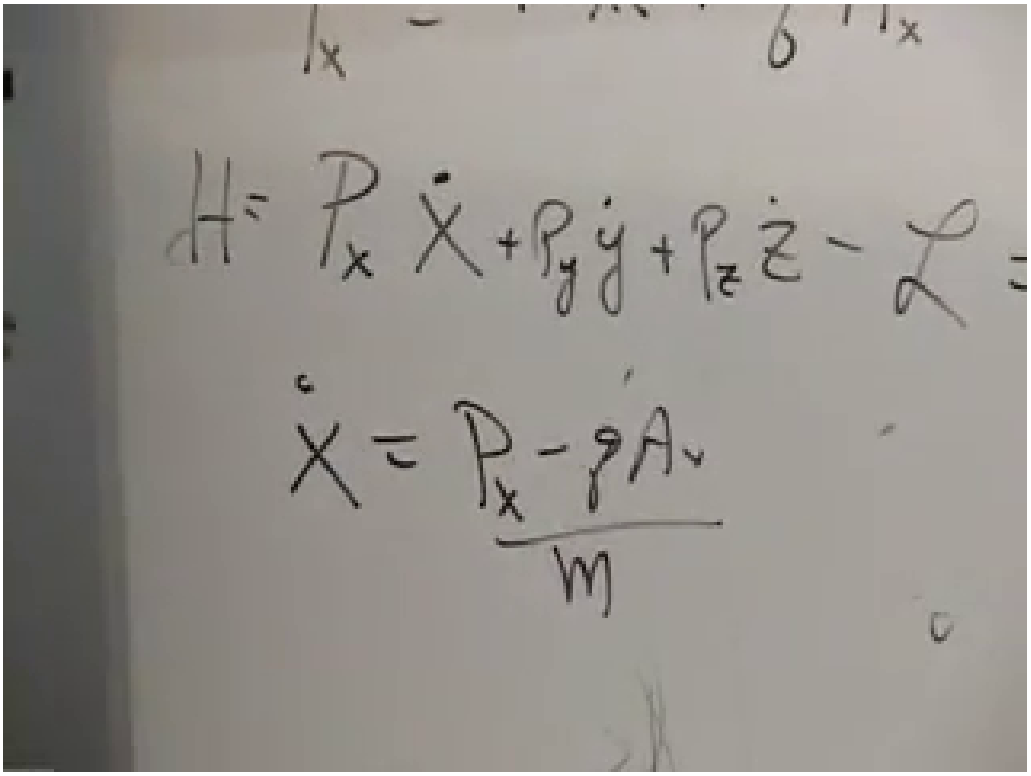

Figure 1.14: The Hamiltonian definition and the inversion $\dot{x} = (p_x - qA_x)/m$ at 01:57:00.

As a final check, Hamilton's equations give

$$\dot{x}_i = \frac{\partial H}{\partial p_i} = \frac{p_i - qA_i}{m} = v_i, \tag{1.37}$$

and

$$\dot{p}_i = -\frac{\partial H}{\partial x_i} = qv_j \frac{\partial A_j}{\partial x_i}. \tag{1.38}$$

Differentiating the mechanical momentum gives

$$\begin{aligned} m\dot{v}_i &= \dot{p}_i - q\frac{dA_i}{dt} \\ &= qv_j \left(\frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j} \right) = q(\boldsymbol{v} \times \boldsymbol{B})_i. \end{aligned} \tag{1.39}$$

Thus Newtonian, Lagrangian, and Hamiltonian descriptions all produce the same magnetic motion. The vector potential changes the canonical bookkeeping, but the observable force depends only on its curl.